

## Module 44 - Array Higher Order Methods (reduce Method)

### 1. What is reduce() Method?

The reduce() method is used to reduce all the elements of an array into a single value by applying a function to each element.

#### Example Code:

```
// Sum of Array Elements

let arr = [10, 2, 3, 4, 5];

// 'accumulator' stores the accumulated result.

// '0' is the initial value of the accumulator.

let totalVal1 = arr.reduce((accumulator, cElement, index, array) => {

    return accumulator + cElement;

}, 0);

// 1st => accumulator + cElement => 0 + 10 => 10
// 2nd => accumulator + cElement => 10 + 2 => 12
// 3rd => accumulator + cElement => 12 + 3 => 15
// 4th => accumulator + cElement => 15 + 4 => 19
// 5th => accumulator + cElement => 19 + 5 => 24

console.log(totalVal1);

let employees = [

    {eName : "Oggy", salary : 10000},

    {eName : "Jack", salary : 20000},

    {eName : "Bob", salary : 30000},

    {eName : "Olivia", salary : 40000}

];

// Calculates the total salary of all employees.

let calcToSalary = employees.reduce((acc, cElement) => {

    return acc + cElement.salary;

}, 0);

console.log(calcToSalary);
```